(M/M/1): (∞/FIFO) Single Server With Infinite Capacity	
1.	**A supermarket has a single cashier. During peak hours, customers arrive at a rate of 20 per hour. The average number of customers that can be processed by the cashier is 24 per hour. Calculate** **a) The probability that the cashier is idle** **b) The average number of customers in the queuing system** **c) The average time a customer spends in the system** **d) The average number of customers in the queue** **e) The average time a customer spends in the queue waiting for service**
Sol.	
2.	**Customers arrive at a sales counter manned by a single person according to a Poisson process with a mean rate of 20 per hour. The time required to serve a customer has an exponential distribution with a mean of 100 seconds. Find the average waiting time of a customer.**
Sol.	
3.	**Customers arrive at a watch repair shop according to a Poisson process at a rate of one per every 10 minutes and the service time is exponential with mean 8 minutes. Find the average number of customers in the shop, average waiting time a customer spends in the shop and the average time a customer spends in the queue for service.**
Sol.	

4.	**In a city airport, flights arrive at a rate of 24 flights per day. It is known that the inter-arrival time follows an exponential distribution and the service time distribution is also exponential with an average of 30 minutes. Find the following** **a) The probability that the system will be idle** **b) The mean queue size** **c) The average number of flights in the queue** **d) The probability that the size exceeds 7**
Sol.	
5.	**Suppose that customers arrive at a Poisson rate of one per every 12 minutes and that the service time is exponential at a rate of one service per 8 minutes. What is** **a) The average number of customers in the system** **b) The average time a customer spends in the system**
Sol.	

(M/M/1): (k/FIFO) Single Server With Finite Capacity	
6.	**Patients arrive at a doctor's clinic according to Poisson distribution at a rate of 30 patients per hour. The waiting room does not accommodate more than 9 patients. Examination time per patient is exponential with mean rate of 20 per hour. Find the:** **a) Probability that an arriving patient will not wait** **b) Effective arrival rate** **c) Average number of patients in the clinic** **d) Expected time a patient spends in the clinic** **e) Average number of patients in the queue** **f) Expected waiting time of a patient in the queue**
Sol.	
7.	**A city has a one-person barber shop which can accommodate a maximum of 5 people at a time (4 waiting and 1 getting haircut). On average, customers arrive at the rate of 8 per hour and the barber takes 6 minutes for serving each customer. It is estimated that the arrival process is Poisson and the service time is an exponential random variable. Find:** **a) Percentage of time the barber is idle** **b) Fraction of potential customers who will be turned away** **c) Effective arrival rate of customers for the shop** **d) Expected number of customers in the barber shop** **e) Expected time a person spends in the barber shop** **f) Expected number of customers waiting for a hair-cut** **g) Expected waiting time of a customer in the queue**
Sol.	

8.	**A one-person barber shop has six chairs to accommodate people waiting for a haircut. Assume that customers who arrive when all six chairs are full leave without entering the barber shop. Customers arrive at the average 3 per hour and spend an average 15 minutes in the barber chair. Find** **(i) What is the probability that a customer can get directly into the barber chair upon arrival?** **(ii) What is the expected number of customers waiting for a haircut?** **(iii) What is the effective arrival rate?**
Sol.	
9.	**Trains arrive at the yard every 15 minutes and the service time is 33 minutes. If the line capacity of the yard is limited to 4 trains, then find** **(i) The probability that the yard is empty.** **(ii) The average number of trains in the system.**
(M/M/s): (∞/FIFO) Multiple Server With Infinite Capacity	
10.	**A travel centre has three service counters to receive people who visit to book air tickets. The customers arrive in a Poisson distribution with the average arrival of 100 persons in a 10 hour service day. It has been estimated that the service time follows an exponential distribution. The average service time is 15 minutes. Find the** **a) Expected number of customers in the system** **b) Expected number of customers in the queue** **c) Expected time a customer spends in the system** **d) Expected waiting time for a customer in the queue** **e) Probability that a customer must wait before he gets service.**
sol.	

9.	**A local hospital has 3 doctors for treating patients. The patients arrive at the hospital according to Poisson distribution with an average of 8 per hour. On an average, a doctor takes about 15 minutes to treat each patient and the actual time taken is known to vary approximately exponentially around this average. Find** **a) Traffic intensity of the system** **b) Probability that a customer has to wait for service** **c) Average number of customers waiting in the queue** **d) Average number of customers in the system** **e) Average waiting time a customer spends in the queue** **f) Expected time a customer spends in the system**
Sol.	
10.	**A petrol pump has 4 pumps. The service time follows an exponential distribution with a mean of 6 minutes and cars arrive for service in a Poisson process at the rate of 30 cars per hour. Find the average waiting time in the queue, average time spent in the system and the average number of cars in the system.**
Sol.	

11.	**A bank has three teller counters. The customers to the bank choose a teller counter at random. It is given that the arrivals to the teller counters are Poisson-distributed at the average rate λ and the mean service rate at the teller counters is $\frac{\lambda}{2}$. Find the steady average queue at each counter.**
Sol.	
(M/M/s): (k/FIFO) Multiple Server With Finite Capacity	
12.	**A barber shop has 2 barbers and 3 chairs for customers. Assume that the customers arrive in a Poisson fashion at a rate of 5 per hour and that each barber service customers according to an exponential distribution with mean of 15 minutes. Further if a customer arrives and there are no empty chairs in the shop, he will leave. Find** **(i) The probability that the shop is empty.** **(ii) Expected number of customers in the shop.**
Sol.	

13.	**A car servicing station has 3 stalls where service is offered simultaneously. The cars wait in such a way that when a stall becomes vacant, the car at the head of the line pulls up to it. The station can accommodate at most 4 cars waiting (7 in the station) at one time. The arrival pattern is Poisson with a mean of one car per minute during peak hours. The service time is exponential with mean 6 minutes. Find** **(i) the average number of cars in the queue** **(ii) the average number of cars in the service station during peak hours.** **(iii) the average waiting time of cars in the system** **(iv) the average waiting time of the cars in the queue.**
Sol.	
14.	**A dispensary has 2 doctors and 4 chairs in the waiting room. The patients who arrive at the dispensary leave when all 4 chairs in the waiting room of the dispensary are occupied. It is known that patients arrive at the average rate of 8 per hour and spend an average of 10 minutes for the check-up and medical consultation. The arrival process is Poisson and the service time is an exponential random variable. Find** **(i) Probability that an arriving patient will not wait.** **(ii) Effective arrival rate at the dispensary.** **(iii) Expected number of patients at the queue.** **(iv) Expected number of patients at the dispensary.** **(v) Expected waiting time of the patient at the queue.** **(vi) Expected waiting time of the patient at the dispensary.**
Sol.	

15.	**A beauty salon has 2 barbers and 6 chairs to accommodate waiting customers. Potential customers, who arrive when all the 6 chairs are full, leave the salon immediately. Customers arrive at the salon at the average rate of 10 per hour and spends an average 10 minutes in the barber's chair. The arrival process is Poisson and the service time is exponential. Find** **(i) Probability of no customer in the beauty salon.** **(ii) Effective arrival rate at the beauty salon.** **(iii) Expected number of customers at the queue.** **(iv) Expected waiting time of the customer at the queue.** **(v) Expected number of customers at the beauty salon.** **(vi) Expected time a customer spends at the beauty salon.**
Sol.	
(M/G/1) Queueing System	
16.	**Automatic car wash facility operates with only one bay. Cars arrive according to a Poisson with mean of 4 cars per hour and may wait in the facility's parking lot If the bay is busy. If the service time for all cars is constant and equal to 10 minutes, then find** L_s, L_q, W_s, W_q **.**
Sol.	

17.	**A car-manufacturing plant uses one big crane for loading cars into the truck. Cars arrive for loading by the crane according to a Poisson distribution with a mean of 5 cars per hour. Given that the service time for all cars is constant and equal to 6 minutes, determine** L_s, L_q, W_s, W_q **.**
Sol.	